

Grace Huang

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Education

Washington University in St. Louis

Expected May 2026

Bachelor of Science in Computer Science + Mathematics (GPA: 3.94 / 4.00)

St. Louis, MO

- **Relevant Coursework:** Data Structures + Algorithms, Object-Oriented Programming, Full Stack Development, Artificial Intelligence, Data Science, Analysis of Algorithms, Systems Software, Data Mining, Data Manipulation & Management at Scale, Mobile Application Development, Parallel & Concurrent Programming, Quantum Computing
- **Activities:** Google Developer Group (Treasurer), Association for Computing Machinery (Treasurer), Lunar New Year Festival (Tinikling Performer), Teaching Assistant

Experience

Software Engineer Intern, Mastercard | O'Fallon, MO

June 2025 – Aug 2025

- Designed a modernized data pipeline using Python and Oracle SQL to deliver structured billing feeds from Mastercard's IST Clearing team to the data warehouse, replacing a legacy mainframe process and removing an intermediary processing layer to streamline data flow.
- Resolved 30+ bugs and reduced 5+ hours of technical debt flagged by SonarQube in the Clearing on Behalf application, improving maintainability and code quality.

Undergraduate Research Assistant, Washington University Z-lab | St. Louis, MO

Aug 2024 – Dec 2024

- Built a full-stack GUI using Flask to control and monitor optical fiber alignment hardware in real time, integrating live video feeds, piezo voltage controls, and real-time power/voltage displays, improving experimental usability for lab researchers.
- Automated fiber alignment for PIC testing using the Nelder-Mead algorithm, maximizing power meter output and reducing tuning time from 3 minutes to 10 seconds, achieving approximately 90% improvement.

Software Engineer Intern, Principal Financial Group | Des Moines, IA

May 2024 – Aug 2024

- Assisted in the development, maintenance, and unit testing of 50+ web components for PFG's design system using TypeScript, Jest, and SCSS, ensuring functionality and usability across company-wide applications.

Teaching Assistant, Washington University | St. Louis, MO

Aug 2023 – Jan 2026

- CSE 2407 (Data Structures), CSE 2400 (Logic & Discrete Math), CSE 3302 (OOP)
- Held office hours, guided problem sets, and graded assignments/labs/exams for 300+ students, providing clear and constructive feedback.

Projects

Leftover to Makeover (Recipe Recommendation App) | Kafka, Spark, Snowflake, Python

- Built an interactive web application that recommends ranked recipes from user-provided ingredients using a feasibility score based on ingredient overlap.
- Designed a distributed data pipeline using Kafka, Spark, and Snowflake to process real-time and batch recipe data, implementing ingredient normalization, unit standardization, and a scalable schema for efficient recipe-ingredient querying.

ARCH Ship (Client Project for ARCH Design) | Java, Java Swing, Gradle

- Developed a Java application that automates art packing calculations, parsing CSV specifications for artwork dimensions, materials, and quantities to generate optimal container arrangements tailored to client requirements.
- Reduced processing time by 99%+ by automating manual calculations and minimizing errors through 145+ unit tests.

Spotify Dashboard & Playlist Generator | Angular, TypeScript, Spotify Web API, HTML, CSS

- Developed a web app that authenticates with Spotify, fetches top tracks/artists/genres, and dynamically generates personalized playlists via chained API requests.
- Designed an interactive dashboard with animated transitions, time-range filters, and real-time loading states to visualize user data and enhance usability.

Technical Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, R, Swift, Standard ML, PHP, HTML/CSS

Frameworks & Libraries: React.js, Angular, Flask, Express.js, Pandas, NumPy, OpenCV

Tools & Technologies: Git, Linux, Oracle SQL Developer, VS Code, Eclipse, Jupyter Notebook, Figma, Spark, Kafka, Airflow, Snowflake